Kalash Thakkar

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RESEARCH INTERESTS

Cosmology and the evolution of the early universe; formation of high-redshift galaxies and large-scale structure; the Cosmic Microwave Background and its secondary anisotropies (kSZ/tSZ effects); machine learning methods for astrophysical inference.

EDUCATION

M.Sc. Astronomy Indore, India
Indian Institute of Technology Indore *2025 – Present*
(Current CGPA: 7.75/10); Coursework includes Galactic and Extragalactic Astronomy, Astrostatistics, Cosmology, and Computational Methods in Astronomy.

B.E. Computer Engineering Surat, India
Gujarat Technological University *2020 – 2024*
Graduated with Distinction (CGPA: 7.99/10); Minor Project: Galaxy Classification using Machine-Learning Algorithms.

RESEARCH EXPERIENCE

Master's Thesis - CNN-Based Halo Velocity Reconstruction via the kSZ Effect
Indian Institute of Technology, Indore | Supervisor: Dr. Priyanka Singh *May 2026 – Present*

- Building a 3D CNN pipeline reconstructing galaxy peculiar velocities from density fields; benchmarking against Zel'dovich theory across CAMELS-TNG and 6 public IllustrisTNG simulations.
- Designing a cross-simulation resolution-robustness study, testing generalization across independently-resolved volumes under a controlled protocol.
- Engineering the full data pipeline (HDF5 parsing, Fourier-space smoothing, 3D patch extraction) and CNN training/eval framework in PyTorch.

Remote Research Collaborator - Generalization of GNN for Cosmic Web Classification
Utica University, USA (Remote Collaboration) | Supervisor: Dr. Unnati Shah *Aug 2026 – Present*

- Developing a GNN (EdgeConv/GCN, PyTorch Geometric) to classify galaxies by cosmic web environment from CAMELS catalogs, using ground-truth labels from a first-principles tidal-tensor (T-web) pipeline.
- Designing a cross-simulation benchmark (TNG→SIMBA/Astrid zero-shot transfer) to quantify domain-shift degradation via macro F1, MCC, and confusion-matrix analysis.
- Extending the architecture to graph transformers (TransformerConv/GPSConv) as a comparative baseline under an identical evaluation protocol.

Summer Research Intern - Radiative Transfer Modeling of the Lyman- Forest
Indian Institute of Technology Indore | Supervisor: Dr. Prakash Gaikwad *Apr 2026 – Jun 2026*

- Parsed and validated z=3 hydrodynamic simulation output, extracting density, temperature, velocity, and HI fraction along 5,000 lines of sight.
- Implemented a first-principles radiative transfer pipeline computing Lyman- optical depth via Voigt-profile integration (Faddeeva function), validated against reference spectra.
- Linked transmitted-flux statistics to IGM structure (voids, filaments, absorbers) using the Choudhury (2001) temperature–density framework.

Winter Research Intern - GPR Pipeline for 21cm EoR Signal Extraction
Indian Institute of Technology, Indore | Supervisor: Prof. Abhirup Datta *Dec 2025 – Jan 2026*

- Developed a Gaussian Process Regression pipeline isolating faint 21cm EoR signal from astrophysical foregrounds in simulated SKA-Low data.
- Modeled covariance via hierarchical kernels (RBF, Matérn-5/2, Exponential), mitigating instrumental wedge systematics and mode-mixing.
- Optimized hyperparameters via Gradient Descent and Nested Sampling (PolyChord), validated through 1D/2D power spectrum estimation.

PROFESSIONAL EXPERIENCE

Programming Language Instructor *Mar 2024 – Aug 2024*
Brighter Institute, Surat

- Designed and taught C/C++/Java/Python curricula for student cohorts.

AI/ML Intern *Dec 2023–Jan 2024*
AlgoBrain AI, Surat

- Implemented ML models (regression, KNN, ensemble methods) and basic CI/CD pipelines.

TECHNICAL SKILLS

- **Languages & Tools:** Python (NumPy, SciPy, Matplotlib, Astropy, GPy, Anesthetic, h5py), C, C++, Java, LaTeX, Git, GitLab, Linux.
- **Machine Learning & Research:** Deep Learning / CNNs (PyTorch), Gaussian Processes, Bayesian Inference (PolyChord), Signal Processing, Power Spectrum Estimation, Simulation-Based Inference, Data Visualization.

COMPETITIVE EXAMINATIONS

- **Joint Admission Test for Masters (IIT JAM) 2025** - Physics (National Level)
  Secured All India Rank (AIR) **329** out of ∼13,000 candidates.
- **Graduate Aptitude Test in Engineering (GATE) 2025** - Physics (National Level)
  Secured All India Rank (AIR) **2592** out of ∼20,000 candidates.

CERTIFICATIONS

- **AWS Academy Machine Learning Foundations** — *Amazon Web Services (AWS)*
- **AWS Academy Cloud Foundations** — *Amazon Web Services (AWS)*
- **AWS Academy Data Engineering** — *Amazon Web Services (AWS)*
- **GCI Data Science and AI Program** — *Matsuo-Iwasawa Laboratory, University of Tokyo, Japan (Remote)*

CONFERENCES & WORKSHOPS

- 7th Indian Planetary Science Conference (IPSC), DAASE, IIT Indore (2026) — *Three-day conference organized by the Indian Planetary Science Association (IPSA), held as part of DAASE's 10th-anniversary celebration.*
- Cosmology at Multiwavelength Meeting, DAASE, IIT Indore (2026) — *Two-day SPARC-sponsored meeting in collaboration with the Statistical Inference group, held as part of DAASE's 10th-anniversary celebration.*
- Summer School in Astronomy, St. Xavier's College, Ahmedabad (2024) — *Hands-on training in data analysis and computational astronomy.*